

GitHub Profile & Project Portfolio

Kentaro Vadney

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GitHub Profile

<https://github.com/VadneyK>

Featured Projects & Technical Achievements

1. Blockchain Carbon Credit Tracking System (CarbonSustain)

Production MVP | May 2024 – Dec 2024

- **Tech Stack:** Node.js, FastAPI, XRPL, Angular, TypeScript, PostgreSQL, AWS (Terraform, EC2, S3, CloudFront)
- **Description:** Architected and built a blockchain-based carbon credit tracking platform from ground up as founding engineering team lead
- **Key Achievements:**
 - Real-time transaction validation supporting 3 enterprise pilot programs
 - Zero-downtime deployment pipeline with full CI/CD automation
 - RESTful API architecture with secure authentication flows
 - Scalable AWS infrastructure managed via Infrastructure-as-Code

2. Enterprise Web Application Optimization (Centigrade)

Production System | Apr 2025 – Aug 2025

- **Tech Stack:** FastAPI, React, Tailwind CSS, PostgreSQL, AWS Kendra, CircleCI, GitHub Actions
- **Description:** Complete rewrite and optimization of critical production services
- **Key Achievements:**
 - 60% reduction in database latency through schema redesign and query optimization
 - 50% improvement in Time-to-Interactive via frontend performance engineering
 - AI-powered document search saving 10+ hours/week using AWS Kendra

- Lighthouse accessibility score improved from 30 to 90
- Same-day deployment capability replacing multi-day release cycles

3. Industrial IoT Cybersecurity Platform (Belden)

R&D & Production | Jul 2023 – May 2024

- **Tech Stack:** Python, Java, Kubernetes, REST APIs, Machine Learning
- **Description:** Container security and AI-powered threat detection for industrial control systems
- **Key Achievements:**
 - LLM-powered threat detection expanding coverage by 30%
 - 40% faster Android testing workflow through architectural improvements
 - Container ingress features for Kubernetes-based IoT deployments

4. Byzantine Fault Tolerance Research (UC Berkeley RISE Lab)

Academic Research | 2021-2022

- **Tech Stack:** Distributed Systems, Consensus Protocols
- **Description:** Led 5-person research team investigating consensus protocol optimizations
- **Achievement:** Co-authored peer-reviewed paper on Byzantine fault tolerance improvements

5. Blockchain Infrastructure & Validation (Berkeley Blockchain Accelerator)

Infrastructure & Analysis | 2021-2022

- **Tech Stack:** AWS, Kubernetes, Stellar Blockchain
- **Description:** Deployed and maintained validator nodes for blockchain networks
- **Achievements:**
 - Evaluated 50+ blockchain startups (portfolio companies raised \$650M+)
 - Production validator node deployment on AWS Kubernetes
 - Infrastructure reliability and monitoring

6. Smart Contract & DeFi Development (Astar Network, OkX)

Blockchain Development Internships | 2021-2022

- **Tech Stack:** Rust, Solidity, React, Web3
- **Description:** Contributed to decentralized exchange features and smart contract infrastructure

- **Achievements:**
 - Production smart contracts in Rust and Solidity
 - React-based DEX user interface components
 - Developer documentation and technical guides
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Core Technical Competencies

Production Experience: 2+ years building and optimizing high-performance web services, blockchain infrastructure, and AI systems

Performance Optimization: Proven track record of 50-60% latency improvements and 5x throughput scaling

Full-Stack Proficiency: End-to-end ownership from database schema to frontend UX, infrastructure to deployment

Emerging Technologies: Hands-on experience with LLM integration, blockchain protocols, and distributed systems research

Code Repository: Visit github.com/VadneyK for code samples and project contributions

Additional Context

All projects listed represent production systems or peer-reviewed research, not academic exercises. I've consistently delivered:

- Measurable performance improvements (50-60% latency reductions)
- Scalable architectures (5x throughput increases)
- Enterprise-grade reliability (zero-downtime deployments)
- Cross-functional impact (AI search saving 10+ hours/week)

I'm particularly passionate about ML/AI systems, blockchain infrastructure, and distributed systems architecture, and I'm seeking opportunities to work on challenging problems in these domains.

For questions about any of these projects or to discuss technical implementation details, please reach out via email or LinkedIn.